# CMSC 420: Disjoint Set Data Structures

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**1 Problems** **2**

# 1 Problems

1. Draw a disjoint forest corresponding to the following disjoint subsets. There are many answers.

$$\{1, 5, 8, 10, 3\}, \{4, 6, 9\}, \{2, 7\}$$

2. Suppose a graph $G$ starts with $n$ nodes and no edges. We represent it by a disjoint set forest. Assume we are using path compression and weighted union.

   (a) Assume $n = 2^k$ for some positive integer $k$. Prove by induction on $k$ that it is possible to find a sequence of edges which could be added to the graph and which would result in the disjoint set forest finally ending up as a single tree of height exactly $k$.

   (b) Is it possible to find a sequence of edges which could be added to the graph and which would result in the disjoint set forest ending up as a single tree of height exactly $n - 1$? Why or why not? Your answer does not need to be particularly formal, a general outline will suffice.

3. Suppose Kruskal's Algorithm (see the notes) is run on the following graph:

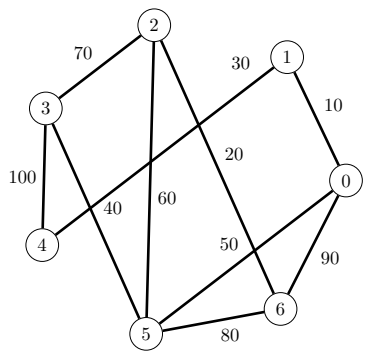


Figure 1: Graph for the Problem

Draw the disjoint forest at the start and then after each edge has been added. Use path compression and weighted union. Your pictures should be after these have occurred.

4. Suppose Kruskal's Algorithm (see the notes) is run on the following graph:

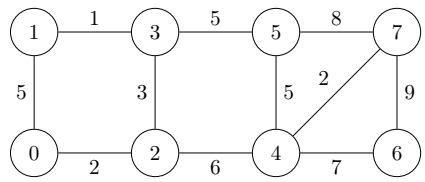


Figure 2: Graph for the Problem

Draw the disjoint forest at the start and then after each edge has been added. Use path compression and weighted union when appropriate. Your pictures should be after these have occurred.

Note: For consistency treat all edges as $x - y$ with $x < y$ and when choices are available choose the edge with smallest $x$. Moreover `union(x,y)` should always be called with $x < y$.

5. A classic way to store a disjoint set forest where the keys are nonnegative integers $0, ..., n-1$ is simply via a list. We create a list $A$ of length $n$ such that for each key $0 \leq k \leq n-1$ the entry $A[k]$ equals the parent key.

   (a) Draw the disjoint set forest represented by the list:

   $$A = [6, 1, 3, 3, 2, 6, 6, 5, 8, 8, 5, 6, 3]$$

   (b) With this storage mechanism write pseudocode for the path-compression version of `findrep` coded as `findrep(A,x)`.

6. Given the following disjoint set forest:

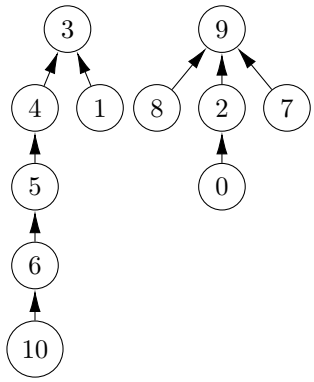


Figure 3: A Disjoint Set Forest

Questions are non-cumulative and assume path compression and weighted union.

(a) Suppose we call `findrep(x)` for $x$ in the left tree and the resulting height of the left tree is unchanged. What could $x$ be?

| | |
|---|---|
| What could $x$ be? | |

(b) Draw the resulting structure when we perform `union(6,0)`.

(c) Suppose we randomly choose an $x$ in the left tree and we randomly choose $y$ in the right tree (all choices equally likely) and we perform `union(x,y)` what is the probability that the resulting tree has each of the following heights. Some have been answered.

| | |
|---|---|
| Height $\leq 1$ | 0 |
| Height 2 | |
| Height 3 | |
| Height 4 | |
| Height $\geq 5$ | 0 |

7. Suppose a 0-indexed list `A` represents a disjoint set forest. Consider the following algorithm:

```
for i = 0 to len(A)-1 inclusive:
    while A[A[i]] != A[i]:
        A[i] = A[A[i]]
    end while
end for
```

What would this algorithm do to the disjoint set forest? Your answer should require at most a sentence or two.

8. Consider the following weighted graph:

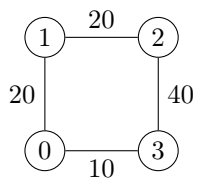


Figure 4: Small Weighted Graph

We run Kruskal's algorithm which iterates three times. What would the corresponding disjoint set forest look like after each iteration?

Note: There are multiple answers, any valid answer will be accepted.

9. Suppose a graph $G$ starts with $n \geq 3$ nodes and no edges. We represent it by a disjoint set forest. Assume we are using path compression and weighted union. Is it possible to find a sequence of edges which could be added to the graph and which would result in the disjoint set forest ending up as a single tree of height exactly $n-1$? Why or why not? Your answer does not need to be particularly formal, a general outline will suffice.

10. Given the following disjoint set forest:

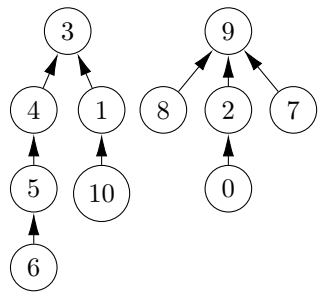


Figure 5: Disjoint Set Forest

Questions are non-cumulative. Assume path compression and weighted union.

(a) Suppose we call `findrep(x)` for some $x$ and the height of at least one tree changes. What could $x$ be?

| | |
|---|---|
| What could $x$ be? | |

(b) Suppose we randomly choose an $x$ in the left tree and we randomly choose $y$ in the right tree (all choices equally likely) and we perform `union(x,y)`. What is the probability that the resulting tree has each of the following heights. Some have been answered.

| | |
|---|---|
| Height $\leq 1$ | 0 |
| Height 2 | |
| Height 3 | |
| Height $\geq 4$ | 0 |